

BES-H₂O₂-Ac

Hydrogen peroxide (H₂O₂) detection probes.

Manufacturer : [Wako](#)

Molecular Formula : C₂₈H₁₁F₇O₈S

M.W. = 640.44 g / mol

Procedure

Step 1: Prepare stock solution (10 mM)

Dissolve 1 mg powder in **156.2 μ L DMSO** to form a 10 mM stock solution. Store at room temperature, must be **protected from light**.

Step 2: Stain

Short time incubation

Dilute to **10 - 50 μ M** with growth medium. Incubate at room temperature for $\approx$ 20 mins, protected from light.

Long time incubation

1. Make medium contains **100 nM** BES-H₂O₂-Ac.
 - Add **4 μ L** of 10 mM stock solution to **400 mL** medium before the medium solidified ($\sim$ 50°C), mix thoroughly, must be protected from light.
2. Grow the plant on this medium for $\approx$ 6 days, cut the root and observe using confocal microscope.

Step 3: Confocal parameters

Set the confocal parameters as follow:

Parameters	Theoretical value	Our machine
Excitation peak	485 nm	488 nm
Emission peak	515 nm	505 ~ 541 nm
Magnification		5X
Laser intensity (488 nm)		20.0%
Master Gain (488 nm)		700 V
Master Gain (T-PMT)		240 V
Digital Gain		1.0
Pinhole		$\approx 35 \mu\text{m}$
Z-stack interval		$16 \mu\text{m}$
Scan speed		6
Scan direction		$\leftrightarrow$

Step 4: Image processing

Open the CZI file with ImageJ, view stack with [Hyperstack](#).

1. Generate BES and T-PMT layers

Repeat this process twice to generate both the BES and T-PMT layers:

[Image](#) → [Stacks](#) → [Z Project...](#) → Projection type: [Max intensity](#) → [File](#) → [Save As](#) → [Tiff...](#)

2. Generate stacks

Load in the previously generated BES and T-PMT TIFF files.

[Image](#) → [Stacks](#) → [Images to Stack](#)

3. Measure distance

Measure the distance between root tip and the BES signal distal boundary. The root tip can be easily recognized in the T-PMT layer.

Done !!!
